MLP (ov4) | 96c4g | numa | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Input + ACK
Credit wait
Output
Host inference
H2D transfer
Torch forward
D2H transfer
Credit send

WG 0 Controller r0

r89 [Worker]
r88 [Worker]
r81 [Worker]
r80 [Worker]
r73 [Worker]
r72 [Worker]
r65 [Worker]
r64 [Worker]
r57 [Worker]
r56 [Worker]
r49 [Worker]
r48 [Worker]
r41 [Worker]
r40 [Worker]
r33 [Worker]
r32 [Worker]
r25 [Worker]
r24 [Worker]
r17 [Worker]
r16 [Worker]
r9 [Worker]
r8 [Worker]
r1 [Worker]
r0 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

WG 1 Controller r2

r91 [Worker]
r90 [Worker]
r83 [Worker]
r82 [Worker]
r75 [Worker]
r74 [Worker]
r67 [Worker]
r66 [Worker]
r59 [Worker]
r58 [Worker]
r51 [Worker]
r50 [Worker]
r43 [Worker]
r42 [Worker]
r35 [Worker]
r34 [Worker]
r27 [Worker]
r26 [Worker]
r19 [Worker]
r18 [Worker]
r11 [Worker]
r10 [Worker]
r3 [Worker]
r2 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

WG 2 Controller r4

r93 [Worker]
r92 [Worker]
r85 [Worker]
r84 [Worker]
r77 [Worker]
r76 [Worker]
r69 [Worker]
r68 [Worker]
r61 [Worker]
r60 [Worker]
r53 [Worker]
r52 [Worker]
r45 [Worker]
r44 [Worker]
r37 [Worker]
r36 [Worker]
r29 [Worker]
r28 [Worker]
r21 [Worker]
r20 [Worker]
r13 [Worker]
r12 [Worker]
r5 [Worker]
r4 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

WG 3 Controller r6

r95 [Worker]
r94 [Worker]
r87 [Worker]
r86 [Worker]
r79 [Worker]
r78 [Worker]
r71 [Worker]
r70 [Worker]
r63 [Worker]
r62 [Worker]
r55 [Worker]
r54 [Worker]
r47 [Worker]
r46 [Worker]
r39 [Worker]
r38 [Worker]
r31 [Worker]
r30 [Worker]
r23 [Worker]
r22 [Worker]
r15 [Worker]
r14 [Worker]
r7 [Worker]
r6 [GPU Ctrl]

Collective gather end
Collective inference end
Collective scatter end

0
40
80
120
160
200
240

ms from each WG's own controller ml_step_start (independent zero; estimated alignment)